Decent Coaching Classes

VIII CBSE

Mind Your Math - IX

Time: 1 Hr **Marks: 30**

SECTION A — Multiple Choice Questions (1 × 6 = 6 marks)

Choose the correct alternative for each of the following questions.

1. Which of the following is a binomial?
   (A) 3a (B) $a^2 + a - 5$ (C) $7ab - 5a$ (D) $5x - 2y + 3xy$
2. Which of the following pairs consists of like terms?
   (A) $2a^2b$ and $6ab^2$ (B) $3x^2y$ and $-5yx^2$ (C) 4xy and $4x^2y$ (D) 7ab and 7a
3. The product $(2x + 3)(x - 4)$ is equal to ___
   (A) $2x^2 - 5x - 12$ (B) $2x^2 + 5x - 12$ (C) $2x^2 - 11x - 12$ (D) $2x^2 - 5x + 12$
4. The factorised form of $4x^2 - 25$ is ___
   (A) $(2x - 5)^2$ (B) $(2x + 5)(2x - 5)$ (C) $(4x + 5)(x - 5)$ (D) $(2x + 25)(2x - 25)$
5. The solution of the equation $5(x - 2) = 3(x + 2)$ is ___
   (A) $x = 2$ (B) $x = 4$ (C) $x = 6$ (D) $x = 8$
6. If $x^2 + kx + 36$ is a perfect square trinomial ($k > 0$), then k = ___
   (A) 6 (B) 12 (C) 18 (D) 36

SECTION B — Very Short Answer Questions (2 × 4 = 8 marks)

7. Write the coefficient of x^2y^2 and the constant term in the expression $7xy + 5x^2y^2 + 7$.
8. Subtract $(4a + 5ab - 7)$ from $(3a - 2ab - 6)$.
9. Factorise: $6x^2y - 9xy^2$
10. Solve: $7x - 3 = 4x + 12$

SECTION C — Short Answer Questions (3 × 4 = 12 marks)

11. Find the product of $3x^2y$ and $(2x - 5y)$, and verify the result for $x = 2$, $y = -1$.

    OR

    Simplify: $6x(x - 3) + x(2 - 5x) + 3x^2$
12. Multiply $(4a + 9)(a - 6)$ and verify the result for $a = 1$.

    OR

    Find the product: $(x - 2y)(3x^2 - 2xy + 2y^2)$
13. Factorise the following:
    (i) $12a^2b - 18ab^2$ (ii) $9a^2 - 30a + 25$ (iii) $2m^2 - 50$
14. Solve: $(x + 2)/3 - (x - 1)/2 = 1/6$ **OR**. Solve: $(3x + 4)/(2 - 6x) = -2/5$

SECTION D — Case-Based Question (4 marks)

15. Mr. Patil has a rectangular vegetable garden. Its length is $(3x + 4)$ metres and its breadth is $(2x - 1)$ metres.
    (i) Write the area of the garden as a polynomial in x. [1]
    (ii) Find the area of the garden when $x = 5$. [1]
    (iii) The perimeter of the garden is 56 m. Form a linear equation in x and solve it to find x. [2]

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